

# This script computes the metrics used in the final text and reporting recipe

```
import numpy as np
import pandas as pd
import geopandas as gpd
import os.path as path
import utils
import matplotlib.pyplot as plt
from collections import Counter

# load all files previously computed
lau_df = utils.load_complex_geojson(path.join(utils.out_data_dir, "drought_days_lau.geojson"))
nuts3_df = utils.load_complex_geojson(path.join(utils.out_data_dir, "nuts3_stats.geojson"))
country_df = pd.read_json(path.join(utils.out_data_dir, "country_stats.geojson"))
```

## Most drought-stricken countries

This outputs all analyzed countries in order, based on how many drought days they experience on average per year.

```
for i, cntr in enumerate(country_df.sort_values(by="median_drought_days", ascending=False)["I"]):
    print(i+1, cntr)
```

```
1 Albania
2 Serbia
3 Poland
4 Liechtenstein
5 North Macedonia
6 Bulgaria
7 Greece
```

- 8 Lithuania
- 9 Switzerland
- 10 Romania
- 11 France
- 12 Croatia
- 13 Czechia
- 14 Slovenia
- 15 Slovakia
- 16 Spain
- 17 Italy
- 18 Cyprus
- 19 Luxembourg
- 20 United Kingdom
- 21 Germany
- 22 Hungary
- 23 Belgium
- 24 Netherlands
- 25 Ireland
- 26 Austria
- 27 Malta
- 28 Portugal
- 29 Denmark
- 30 Sweden
- 31 Finland
- 32 Iceland
- 33 Estonia
- 34 Norway
- 35 Latvia

```
country_df.sort_values(by="max_drought_days")[:5][["NAME_ENGL", "median_drought_days", "max_drought_days"]]
```

|    | NAME_ENGL | median_drought_days | max_drought_days |
|----|-----------|---------------------|------------------|
| NO | Norway    | 34.11               | 69.70            |
| FI | Finland   | 40.98               | 104.12           |
| AT | Austria   | 56.52               | 110.29           |
| SE | Sweden    | 44.94               | 119.68           |
| IS | Iceland   | 35.93               | 148.81           |

Looking at the countries with the lowest “max\_drought\_days” value: -> all countries in Europe have faced drought in the observed timeframe.

## Europe-wide statistics

This section splits the nuts3-dataset along a threshold for median and maximum drought days. From this it computes a portion of surface area

```
drought_days_median_50_mask = nuts3_df["median_drought_days"] > 50
dry_nuts3_df = nuts3_df[drought_days_median_50_mask]
wet_nuts3_df = nuts3_df[~drought_days_median_50_mask]
dryland_km2, wetland_km2 = dry_nuts3_df["area_km2"].sum(), wet_nuts3_df["area_km2"].sum()
dryland_km2 / (dryland_km2+wetland_km2)
```

```
np.float64(0.6541255404617256)
```

-> almost two thirds of the EEA+UK area experiences on average more than 50 days of drought warnings per year.

```
drought_days_median_100_mask = nuts3_df["median_drought_days"] > 100
dry_nuts3_df = nuts3_df[drought_days_median_100_mask]
wet_nuts3_df = nuts3_df[~drought_days_median_100_mask]
dryland_km2, wetland_km2 = dry_nuts3_df["area_km2"].sum(), wet_nuts3_df["area_km2"].sum()
dryland_km2 / (dryland_km2+wetland_km2)
```

```
np.float64(0.11943094078060766)
```

-> Over 10% of the EEA+UK area experiences on average more than 100 days of drought warnings per year.

```
drought_days_max_183_mask = nuts3_df["max_drought_days"] >= 183
dry_nuts3_df = nuts3_df[drought_days_max_183_mask]
wet_nuts3_df = nuts3_df[~drought_days_max_183_mask]
dryland_km2, wetland_km2 = dry_nuts3_df["area_km2"].sum(), wet_nuts3_df["area_km2"].sum()
dryland_km2 / (dryland_km2+wetland_km2)
```

```
np.float64(0.6648711964996835)
```

-> Two thirds of all regions in EEA+UK have experienced a severe drought year since 2012, in which a drought warning was issued for over 183 Days (half a year).

## NUTS-3-regions with the highest median number of drought days

```
nuts3_df.sort_values(by="median_drought_days", ascending=False)[:10][["nuts_id", "cntr_code"]
```

|      | nuts_id | cntr_code | name_latn        | median_drought_days | max_drought_days |
|------|---------|-----------|------------------|---------------------|------------------|
| 8    | AL032   | AL        | Fier             | 199.85              | 334.84           |
| 1110 | PL923   | PL        | Płocki           | 178.71              | 345.02           |
| 1099 | PL822   | PL        | Przemyski        | 175.58              | 253.82           |
| 128  | CH032   | CH        | Basel-Landschaft | 167.22              | 281.20           |
| 670  | ES611   | ES        | Almería          | 165.54              | 307.81           |
| 1279 | UKE32   | UK        | Sheffield        | 165.23              | 279.97           |
| 11   | AL035   | AL        | Vlorë            | 162.00              | 267.75           |
| 609  | EL542   | EL        | Thesprotia       | 159.92              | 309.65           |
| 983  | MK005   | MK        | Pelagoniski      | 156.90              | 306.41           |
| 607  | EL533   | EL        | Florina          | 156.30              | 273.65           |

```
lau_df.sort_values(by="median_drought_days", ascending=False)[:10][["GISCO_ID", "CNTR_CODE",
```

|       | GISCO_ID    | CNTR_CODE | LAU_NAME                 | median_drought_days | max_drought_d |
|-------|-------------|-----------|--------------------------|---------------------|---------------|
| 701   | ES_18056    | ES        | Cúllar                   | 280.94              | 357.73        |
| 18662 | EL_41010301 | EL        | K A T                    | 275.00              | 340.00        |
| 18670 | EL_41010309 | EL        | K II                     | 275.00              | 340.00        |
| 2237  | ES_18029    | ES        | Benamaurel               | 270.95              | 359.75        |
| 2742  | ES_18187    | ES        | Villanueva de las Torres | 263.94              | 360.00        |
| 18665 | EL_41010304 | EL        | K A                      | 262.90              | 349.54        |
| 18669 | EL_41010308 | EL        | K II                     | 252.42              | 346.28        |
| 707   | ES_18064    | ES        | Dehesas de Guadix        | 250.94              | 352.96        |
| 15048 | EL_10020117 | EL        | K T                      | 248.38              | 314.28        |
| 18666 | EL_41010305 | EL        | K H                      | 246.38              | 348.17        |

```
avg_drought_days_gt_100 = len(lau_df[lau_df["median_drought_days"] > 100])
print(f"{avg_drought_days_gt_100} of {len(lau_df)} LAUs ({100* avg_drought_days_gt_100 / len
```

19362 of 99114 LAUs (19.54%) experience more than 100 drought days per year.

```

max_drought_days_gt_200 = len(lau_df[lau_df["max_drought_days"] > 200])
print(f"{max_drought_days_gt_200} of {len(lau_df)} LAUs ({100* max_drought_days_gt_200 / len(lau_df)}%) experienced at least one year between 2012 and 2025 with over 200 days of drought")

print("most frequent 'worst year' at LAU-scale:")
print(lau_df["max_drought_days_year"].value_counts()[:5])

```

82779 of 99114 LAUs (83.52%) experienced at least one year between 2012 and 2025 with over 200 days of drought.  
most frequent 'worst year' at LAU-scale:

```

max_drought_days_year
2019    20158
2022    19760
2017    13365
2012    13045
2018     7216
Name: count, dtype: int64

```

```

nuts3_df["cropland_km2_by_type"] = nuts3_df["cropland_km2_by_type"].apply(lambda x: eval(x))
high_drought_df = nuts3_df[nuts3_df["median_drought_days"] > 100]
low_drought_df = nuts3_df[nuts3_df["median_drought_days"] <= 100]

```

```

low_drought_crops = sum([Counter(dct) for dct in low_drought_df["cropland_km2_by_type"].values])
high_drought_crops = sum([Counter(dct) for dct in high_drought_df["cropland_km2_by_type"].values])

```

```

sum(low_drought_crops.values()) + sum(high_drought_crops.values())

```

0

```

# 1. Align keys (intersection is safest)
keys = sorted(set(low_drought_crops) & set(high_drought_crops))

# 2. Compute ratio (avoid division by zero)
def safe_ratio(k):
    return low_drought_crops[k] / high_drought_crops[k] if high_drought_crops[k] != 0 else float('inf')

# 3. Sort keys by ratio
keys_sorted = sorted(keys, key=safe_ratio)

# 4. Extract values in sorted order
vals1 = np.array([low_drought_crops[k] for k in keys_sorted])
vals2 = np.array([high_drought_crops[k] for k in keys_sorted])

```

```
# 5. Plot stacked bar chart
x = np.arange(len(keys_sorted))

plt.figure()
plt.bar(x, vals1, label='low drought LAUs (<=100 days)')
plt.bar(x, vals2, bottom=vals1, label='high drought LAUs (>100 days)')

plt.xticks(x, keys_sorted, rotation=45, ha="right")
plt.legend()
plt.xlabel('crop type')
plt.ylabel('area (km2)')
plt.title('cropland area by type and drought propensity')

plt.show()
```

